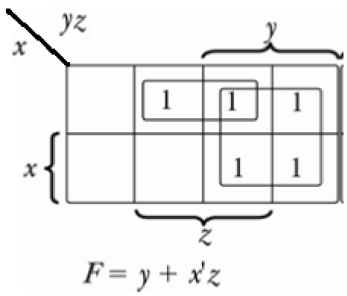
Homework 4 Solutions – Spring 2021

1. Problem 1-8 (b)
   Simplify the following Boolean functions using three-variable maps.

   $$F(x,y,z) = \Sigma\,(1,2,3,6,7)$$

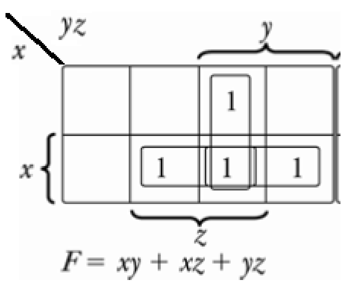
   Solution:



2. Problem 1-8 (c)
   Simplify the following Boolean function using three-variable maps.

   $$F(x,y,z) = \Sigma\,(3,5,6,7)$$

   Solution:

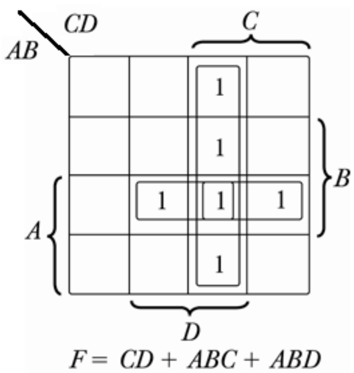


3. Problem 1-9 (b)

   Simplify the following Boolean function using four-variable maps.

   $$F(A,B,C,D) = \Sigma\,(3,7,11,13,14,15)$$

   Solution:

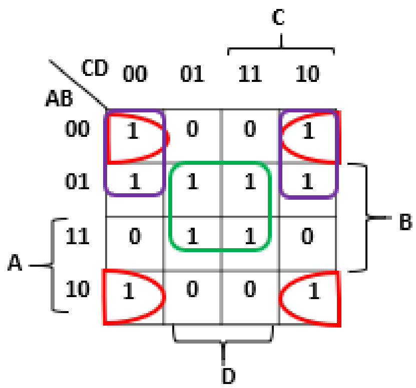


4. Problem 1-9 (d)

   Simplify the following Boolean functions using four-variable maps.

   $$F(A,B,C,D) = \Sigma\,(0,2,4,5,6,7,8,10,13,15)$$

Solution:



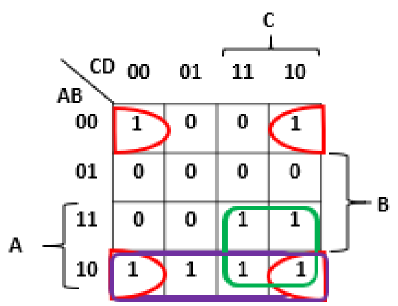
F =A'D' + B'D' + BD

5. Problem 1-11

   Simplify the following Boolean function in sum-of-products form by means of a four-variable map. Draw the logic diagram with (a)AND-OR gates;(b) NAND gates.

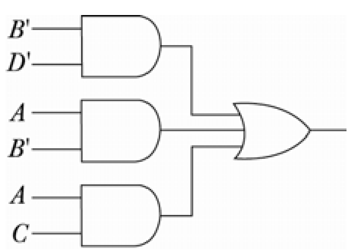
$$F(A,B,C,D) = \Sigma\ (0,2,8,9,10,11,14,15)$$

   Solution:

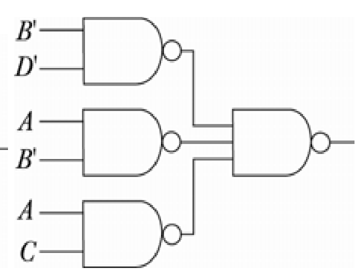


$$F = B'D' + AB' + AC$$

(a)



(b)



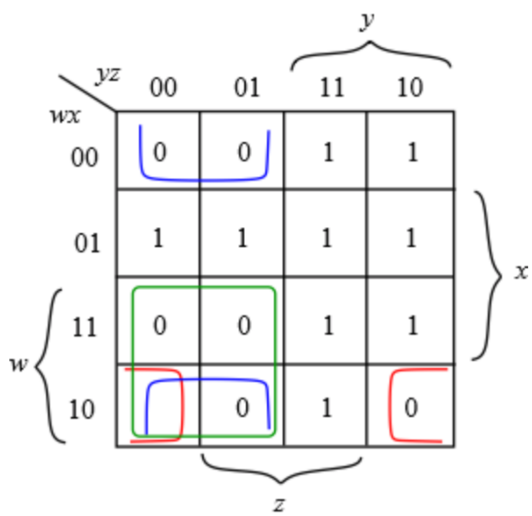
6. Problem 1-12

   Simplify the following Boolean function in product-of-sums form by means of a four-variable map. Draw the logic diagram with (a) OR-AND gates; (b) NOR gates.

$$F(w,x,y,z) = \Sigma\ (2,3,4,5,6,7,11,14,15)$$

Solution:

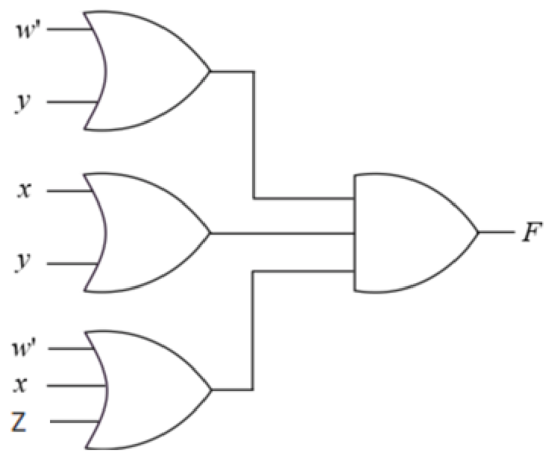
(a)



$$(F')' = (wy' + x'y' + wx'z')'$$

$$F = (wy')'(x'y')'(wx'z')'$$

$$F = (w' + y)(x + y)(w' + x + z)$$



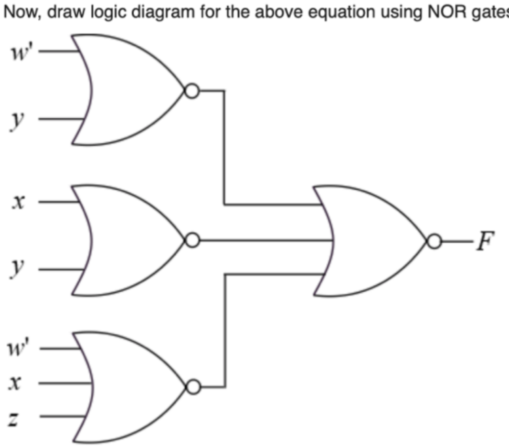
(b)

$$F = (w' + y)(x + y)(w' + x + z)$$

The above can be written as follows:

$$\begin{aligned} F &= \left(\left((w' + y)(x + y)(w' + x + z)\right)'\right)' \\ &= \left((w' + y)' + (x + y)' + (w' + x + z)'\right)' \end{aligned}$$

Now, draw logic diagram for the above equation using NOR gates , as follows:



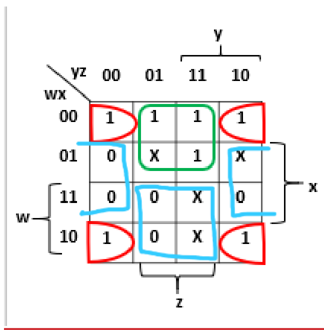
7. Problem 1-13

Simplify the Boolean function *F* together with the don't-care conditions *d* in (1) sum-of-products form (2) product-of-sums form.

$$F(w,x,y,z) = \Sigma\,(0,1,2,3,7,8,10)$$

$$F(w,x,y,z) = \Sigma\,(5,6,11,15)$$

Solution:

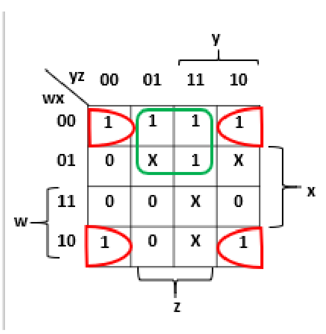


(1) F = x'z' + w'z [solved from values grouped in 'red and green' color]

F = (x' + z) (w' + z') [solved from values grouped in 'blue' color]. ~~F = x'z' + w'z~~

(2)

~~(1)~~ ~~F = (x' + z) (w' + z')~~



8. Simplify the following Boolean expression using two-variable Karnaugh maps: (Hint: Write a truth table for the function and then use Karnaugh map to simplify)
   a. F(x, y) = x'y' + yy + x'yy'
   b. F (x, y) = xy + x′y′y′ + x′yy′

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Solution: a) Truth table:

x	**y**	**F**
0	0	1
0	1	1
1	0	0
1	1	1

X \ Y	**0**	**1**
0	1	1
1	0	1

Answer: x'+y

b) F (x, y) = xy + x′y′y′ + x′yy′

Truth table:

x	**y**	**F**
0	0	1
0	1	0
1	0	0
1	1	1

X \ Y	**0**	**1**
0	1	0
1	0	1

Answer: x'y'+xy

9. Simplify the following Boolean expression using three-variable Karnaugh maps: (Hint: Write a truth table for the function and then use Karnaugh map to simplify)
   a. F(A, B, C) = A'BC + A'BC' + AB'C' + AB'C
   b. F(A, B, C) = A'B + BC' + B'C'

Solution: a) Truth table:

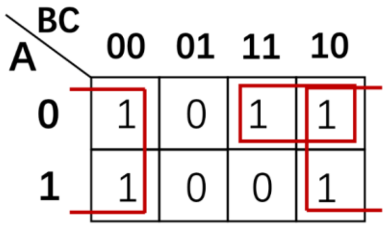
A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

A \ BC	00	01	11	10
0	0	0	1	1
1	1	1	0	0

Answer: F (A, B, C) = A′B + AB′

b) F(A, B, C) = A'B + BC' + B'C'

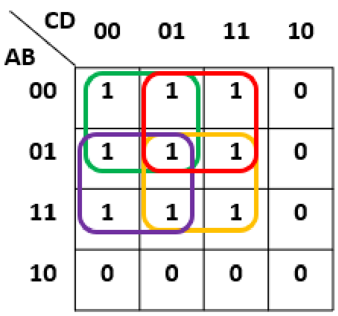
A	**B**	**C**	**F**
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0



Answer: C’ + A’B

10. Simplify the following Boolean expression using four-variable Karnaugh maps:
F (A, B, C, D) = A’B’C’D’ + A’B’C’D + A’B’CD + A’BC’D’ + A’BC’D +A’BCD + ABC’D’ + ABC’D + ABCD

Solution:

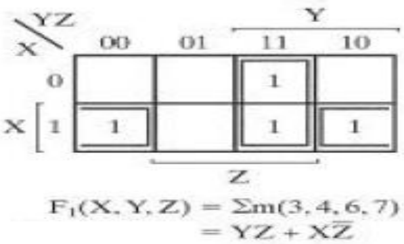


F (A, B, C, D) = A’C’ + A’D + BC’ + BD

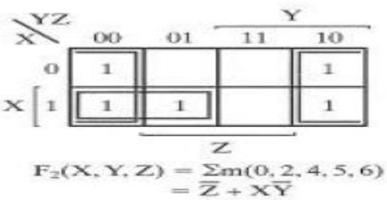
11. Simplify the following Boolean functions using Karnaugh map
    a. F(X, Y, Z) = ∑m(3,4,6,7)
    b. F(X, Y, Z) = ∑m(0,2,4,5,6)
    c. F(W, X, Y, Z) = ∑m(0,1,2,4,5, 6,8,9,12,13,14)
    d. F(A, B, C, D) = ∑m(l,3,7,11,15) and d(A,B,C,D) = ∑m(0,2,5)
    e. F(w, x, y, z)= Σm(1,4,5,6,12,14,15)

Solution:

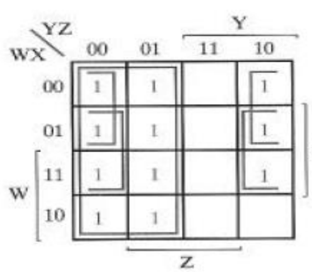
a.



b.

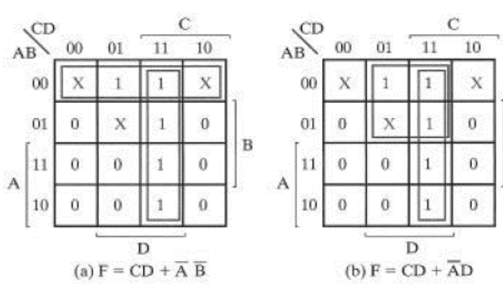


c.

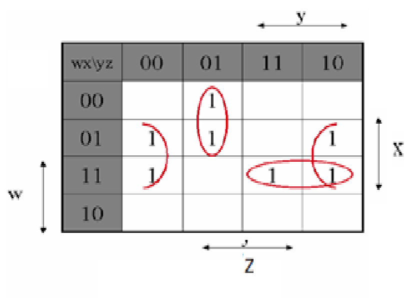


$$F = \overline{Y} + \overline{W}\,\overline{Z} + X\overline{Z}$$

d.



e.

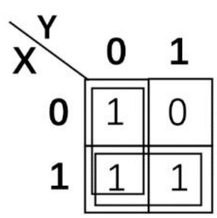


w′y′z + xz′ + wxy

12. Simplify the following expressions in (1) sum of products and (2) product of sums:
    (Hint: Write a truth table for the function and then use Karnaugh map to simplify)
    a. F(x, y) = x'y' + y'y' + yy' + xy
    b. F(A, B, C, D) = (A′ + B′ + D′) (A + B′ + C′) (A′+ B + D′)(B+ C′ + D′)

Solution:

(a) Method 1: Using truth table below, we get the **sum of product** is F = x + y’ (where the output is 1), and the **product of sum** is F = x + y’ (where the output is 0).

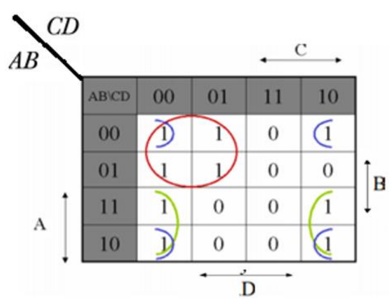


Method 2: We use the F function in sum of products, then we have following truth table. The **sum of product** is F = y’+x (where the output is 1), and the **product of sum** is F = y’ + x (where the output is 0).

X	Y	F
0	0	1
0	1	0
1	0	1
1	1	1

(b) F(A, B, C, D) = (A′ + B′ + D′) (A + B′ + C′) (A′+ B + D′)(B+ C′ + D′)

A	B	C	D	F
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	1
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	1
1	1	1	1	0



Sum of product: B’D’+AD’+A’C’

Product of sum: (A’+D’)(C’+ D’)(A+B’+C’)